

Research Statement

Nahee Kim

Humans begin learning language from the moment they are born by observing the sounds, expressions, and actions around them. Although young children cannot explicitly explain the grammar of the language they use, they gradually learn the relationships between recurring expressions and the situations in which they appear. Through these relationships, they become able to understand and produce appropriate sentences. Yet understanding language involves more than knowing the meanings of individual words. The same expression may require entirely different responses depending on who says it, when it is said, what has happened previously, and what the speaker intends. Language is ultimately a means of guiding action, and acting appropriately requires an understanding of the context in which an utterance is situated.

I experienced university life across the period before and after the emergence of ChatGPT. When using early language models, the greatest difficulty was that they could generate grammatically natural sentences while misunderstanding what the user had actually requested and producing irrelevant answers. At the time, I regarded this merely as an inconvenience caused by insufficient model performance. However, as language models become connected to search engines, external documents, personal information, and tools, and evolve into agents capable of carrying out actions autonomously, misunderstanding context no longer remains a minor response error. If a model cannot distinguish a user's instruction from a sentence contained in external data, or if it gradually deviates from the original objective as an interaction progresses, it may perform unintended actions and create serious security risks.

My interest in this problem is also connected to a personal experience. While attempting to purchase a pair of AirPods through an online second-hand marketplace, I was deceived and lost approximately \$135. None of the individual messages appeared obviously suspicious at the time. Each request seemed plausible in isolation, and each new explanation appeared to justify the previous one. Only afterward did I recognize that the interaction had gradually redirected my decisions away from the original purpose of the transaction. The problem was not simply that one sentence was false. Rather, I failed to notice that the evolving interaction no longer fit the broader context and objective of the purchase.

This experience led me to view deception as a problem of contextual inconsistency and gradual redirection. Humans often recognize that something is wrong before they can identify a single clearly false statement; they notice that a request, explanation, or action does not fit the history, intention, or structure of the interaction. A similar problem arises in AI systems. While carrying out a user's request, an agent may encounter an instruction or piece of external information that appears reasonable in isolation but subtly conflicts with the user's original objective. In this sense, adversarial information can redirect an AI system much as a deceptive message can redirect a human decision. If a model can represent the evolving context and purpose of an interaction, it may also be able to identify such goal-inconsistent information before it is incorporated into its decision-making process.

These observations motivate the questions that guide my research: How do AI models internally distinguish instructions, evidence, and irrelevant or adversarial information while forming context over time? How are these relationships represented and transformed in hidden-state space? Can changes in internal trajectories reveal when a model begins to depart from the user's intention? Finally, can these trajectories be redirected before contextual deviation results in an unintended or unsafe action?

1

Research Conducted During My Master's Program

1.1 TST-Mamba (Submitted to Neurocomputing, under review)

NLOS Transient Data as a Physical Language and Mamba-Based Selective Signal Learning

When I began conducting research in a computer vision laboratory, I came to view NLOS transient data as another form of language that a model must interpret. In non-line-of-sight (NLOS) imaging, a camera cannot directly ob-

serve the target scene. Instead, the hidden scene must be inferred from the arrival times and intensities of light reflected between a visible wall and a concealed object. Each measurement is noisy and ambiguous in isolation, yet signals distributed across time and space indirectly encode the structure of the hidden scene. Just as language is interpreted within the context of a conversation, a transient signal acquires physical meaning through its surrounding spatiotemporal relationships.

I did not regard this task as a simple denoising problem. The central challenge was to distinguish signals required for reconstruction from irrelevant disturbances while preserving the fine structures needed to infer the hidden scene. To address this challenge, I focused on Mamba, a state-space model capable of efficiently learning long-range sequential dependencies.

Mamba processes inputs sequentially, accumulates prior information in a hidden state, retains information relevant to the current input, and suppresses what is unnecessary. I interpreted this mechanism as a process of context formation rather than the independent processing of individual measurements. In NLOS transient data, a single time bin cannot reveal whether a measurement is a valid photon response or background noise. This requires jointly considering nearby temporal changes and signals from neighboring spatial locations. I therefore used Mamba to scan the temporal sequence at each spatial location and treated the final hidden state as a compressed bottleneck representation in which information from the full interval had been selectively accumulated. Spatial Mamba then processed these bottleneck tokens so that the model could learn how signals at different locations relate to one another. Temporal Mamba thus forms context over time, while Spatial Mamba integrates context across space.

However, the compressed representation alone could not preserve every sharp temporal peak and weak photon return required for reconstruction. I therefore transferred the full temporal features through a skip path and combined them with the bottleneck code containing spatial context. The bottleneck path provides the global relationships needed to interpret the signal, whereas the skip path preserves local evidence that may be lost during compression. Combining these paths allows the model to reconstruct a clean transient using both accumulated context and fine-grained observations.

Through this analysis, I formalized TST-Mamba as a Mamba-based skip-connected denoising autoencoder. Temporal Mamba encodes the noisy transient sequence into a compressed representation, Spatial Mamba forms spatial context in latent space, and the skip-latent fusion decoder combines this context with the original temporal details. This work showed that Mamba can be interpreted not merely as an efficient long-sequence architecture, but as a mechanism that learns what to remember, what to suppress, and which information to use for its final judgment.

Through this research, I came to believe that “context” does not simply mean providing a model with more information. It involves selecting task-relevant information, relating observations separated in time or space, suppressing irrelevant disturbances, and preserving evidence needed for the final decision. In NLOS transients, an incorrect selection may cause the model to confuse signal with noise or remove weak but valid responses. Similarly, in language models, misinterpreting the role or source of information may blur the distinction between user instructions and external data and cause behavior to deviate from the intended task. This experience expanded my interest beyond physical-signal restoration toward understanding how context is formed and transformed within AI models and how those internal changes influence their decisions and behavior.

1.2 RiemannMamba (Under review, NeurIPS 2026)

A Geometric Interpretation of Mamba’s Internal Representations

Through the TST-Mamba project, I became interested in how Mamba selectively retains and suppresses information while forming task-relevant context in its hidden state. This led me to ask a further question: As Mamba processes an input sequence, which tokens induce lasting changes in the internal state, and might different tokens possess different “durations” and degrees of influence even when they are processed in the same order?

The final output alone cannot reveal which information was quickly forgotten and which continued to affect later hidden-state evolution. Existing explainable AI methods primarily estimate how individual input features contribute to the final prediction. In a state-space model such as Mamba, however, it is also necessary to understand when information is incorporated, how long its influence persists, and how it changes through interactions with

subsequent inputs. I therefore viewed hidden-state evolution not as a collection of independent points in a high-dimensional space, but as geometric trajectories that may progress at different rates for different tokens.

This perspective was inspired by the concept of proper time in Einstein’s general theory of relativity. Proper time is not an absolute quantity shared identically by all observers; it depends on the geometry of spacetime and the path followed by each observer. This led me to ask whether tokens processed by Mamba might similarly possess an “internal time” distinct from their externally assigned order. Although all tokens pass through the same network depth and scan sequence, their content and surrounding context differ. Some may induce only small hidden-state changes, whereas ambiguous or complex inputs may cause larger changes that persist during subsequent processing. Thus, identical computational steps may correspond to different accumulated trajectories in representation space.

This idea connects naturally to Mamba’s input-dependent step size Δ , which controls the degree of state transition induced by each input. Rather than interpreting Δ only as a computational gate, I considered it as a geometric scale that determines how hidden-state changes should be measured. I treated the token-wise hidden-state space as a Riemannian manifold and constructed state-dependent metric weights from Δ . Hidden-state changes before and after the selective scan were then measured as locally weighted line elements rather than simple Euclidean distances. By accumulating these changes across layers, I defined the geometry-weighted path length traveled by the model while processing an input as its internal time τ . Here, τ represents neither token order nor actual computation time, but the accumulated hidden-state evolution induced by the input.

I hypothesized that easy inputs would produce relatively simple trajectories and shorter internal times, whereas difficult or error-prone inputs would induce larger and more complex state changes. In practice, RiemannMamba produced larger τ values for misclassified, corrupted, and distribution-shifted inputs, showing that inputs processed through the same computational depth may nevertheless induce substantially different internal dynamics. These results suggest that internal-state evolution can provide a reliability signal distinct from final-output confidence. In addition, Tau-CAM, which visualizes the spatial distribution of internal time, revealed which input regions produced large changes in the model’s internal representations.

1.3 Driver Handover Readiness Estimation via Bio–Vehicle Signal Fusion and Safety–Oriented Re-decision (Accepted to the IEIE Summer Annual Conference 2026)

Safe Decision-Making in Human–Machine Collaboration

In conditional automated driving, transferring control of the vehicle back to the driver is not merely a driver-state classification problem, but a decision-making problem that must safely coordinate collaboration between a human and an automated system. Before handing control to the driver, the vehicle must determine whether the driver has recognized the current driving situation and is actually ready to resume driving. Transferring control too early to an unprepared driver increases the risk of an accident, while excessively delaying a necessary handover may also compromise safety.

I conducted a multimodal learning study that estimated driver handover readiness by jointly using the driver’s physiological signals and vehicle signals. Physiological signals reflect the driver’s cognitive and physical arousal and readiness, whereas vehicle signals describe the current dynamic state of the vehicle, including speed, steering, acceleration, and deceleration. Because the two modalities provide information about different aspects of the situation, neither alone is sufficient to determine whether a safe handover is possible. I therefore integrated the two signal types to evaluate the states of the driver and vehicle together and to achieve more stable readiness estimation than would be possible with a single modality. However, high classification accuracy alone cannot guarantee safety in real driving environments. The driver’s condition and the vehicle’s driving state continue to change, meaning that a handover decision that was appropriate at the initial decision point may later become unsafe. I therefore studied a safety-oriented re-decision structure that does not fix the initial readiness estimate as the final decision, but instead reassesses handover feasibility using newly observed physiological and vehicle signals. When the model’s prediction is uncertain or the driver’s readiness is judged insufficient, the system withholds or reevaluates the transfer of control, preventing a single erroneous prediction from immediately leading to a hazardous action. This research taught me that the role of AI in systems where humans and machines collaborate is not simply to replace the human, but to coordinate when and how actions should be taken while jointly considering the human’s

condition and the system’s limitations. In safety-critical environments, the ability to recognize uncertainty and revise an earlier decision in response to a changed state is more important than trusting a single prediction. This experience led me to view trustworthy AI not simply as a high-accuracy model, but as a system that can collaborate with people while preserving human control and safety.

1.4 Integration of My Research Experience

TST-Mamba examined how a model selects and preserves information in spatiotemporal data, while Riemann-Mamba investigated how the selected information shapes hidden-state dynamics and how those dynamics relate to task difficulty and reliability. The Driver Handover project extended this perspective from internal representations to action, showing that a decision must be reevaluated when the state of the human or the environment changes. Together, these studies led me to view trustworthy AI as a process rather than a property of a single output: reliable behavior depends on how information is selected, how internal states evolve, and whether a system can revise its decision as the surrounding context changes.

Building on this integrated perspective, my future research will focus on identifying early signs of task or intention drift from hidden-state trajectories and on testing whether targeted interventions can redirect those trajectories before they lead to unreliable or unsafe behavior.

2 Future Research

Geometry-Aware White-Box Analysis and Control of AI Systems

The most fundamental question I want to investigate is: What actually happens inside an AI model as it forms, revises, or abandons an interpretation? My future research will examine this question through the geometry of evolving hidden representations. The proposed framework is not limited to a single architecture. In state-space models, hidden states form explicit sequential trajectories over time, while in Transformer-based models, representations can be examined as trajectories that evolve across layers, tokens, and interaction turns. I will use state-space models as a tractable setting for developing this framework, while examining its relevance to language models and agentic systems from the early stages of the research.

My goal is to develop white-box methods that reveal not only which input features influenced a final prediction, but also how instructions, evidence, and external information were internally distinguished and combined during decision formation. This distinction becomes a security problem when a model processes both trusted user instructions and untrusted external data. An adversary may embed an instruction within a document, message, webpage, or tool output so that the model gradually abandons the user’s original objective while its final response remains apparently plausible. Building on prior work that formulates such instruction-induced deviation as Task Drift and detects it through changes in model activations [1], I aim to determine whether manipulation produces detectable changes in hidden-state geometry and whether those changes can support intervention before the model performs an unsafe action.

2.1 Hidden-State Geometry as a Basis for White-Box AI

Deep learning models represent hidden states as vectors $h \in \mathbb{R}^d$, and their internal changes are often analyzed using Euclidean notions of distance and direction. However, representing a state with Euclidean coordinates does not require the meaningful geometry of the representation space to be globally flat. Language, images, time series, and physical signals contain nonlinear and structured relationships that may be better described through a curved latent manifold.

I therefore hypothesize that model computation involves two coupled processes: the evolution of the hidden state and the evolution of the geometry through which that state is interpreted,

$$h_{k+1} = F(h_k, x_k), \quad g_k = G(h_k, x_k),$$

where g_k denotes a local, input-dependent metric and k may index time steps, layers, tokens, or interaction turns. Under this view, an input does not merely move a representation through a fixed space. It may also alter the local geometry that determines which directions are important, which states remain close, and which relationships continue to influence later computation.

This perspective is formally inspired by the relationship between matter and geometry in general relativity. I do not claim that neural networks literally obey physical laws. Rather, I draw on the principle that the conditions within a system may reshape the geometry through which its dynamics unfold. Analogously, information and accumulated context may reshape latent geometry, while hidden states evolve along the resulting paths.

This framework also suggests a geometric interpretation of memory. Long-range dependencies may rely not only on retaining earlier state values or providing later access to token representations, but also on forming a geometry in which contextually related states remain effectively close. States that are distant in sequence order or Euclidean coordinates may have a short task-dependent geodesic distance. From this perspective, useful representation learning involves learning not only useful coordinates, but also a useful geometry that organizes internal relationships and trajectories.

A geometry-weighted trajectory length can be expressed conceptually as

$$\tau = \sum_k \sqrt{\Delta h_k^\top g_k \Delta h_k}, \quad \Delta h_k = h_{k+1} - h_k.$$

Here, τ is not physical time. It represents the accumulated amount of internal change measured under the geometry induced by the input and context. Together with curvature, changes in direction, and geodesic relationships, this quantity may provide an interpretable description of how a model forms, preserves, and revises its internal context.

Rather than examining only the magnitude of an activation change at a single point, I aim to study how internal changes accumulate across an entire trajectory, where divergence begins, and how the relationships among internal states are reorganized. This may make it possible to localize when and where external information begins to reshape the model’s interpretation of its original task.

2.2 Research Questions under Adversarial Context Manipulation

I will organize this research around three experimentally falsifiable questions.

Q1. Does input-adaptive geometry reveal task drift [1] more clearly than conventional internal-state measurements?

I hypothesize that clean task execution, benign context variation, and instruction injection produce distinguishable hidden-state trajectories. I will examine whether geometry-aware quantities such as trajectory length, curvature, changes in direction, and local metric variation provide information beyond Euclidean activation differences, internal probes, output confidence, and output-level safety checks. This hypothesis would not be supported if the geometric description provides no consistent advantage in detecting or localizing task drift across architectures and attack conditions.

Q2. Are geometric drift signals robust to an adaptive attacker?

A detector that succeeds only against fixed or previously observed injections would be insufficient as a security mechanism. I will therefore consider an attacker who is aware that internal trajectories are being monitored and attempts to alter model behavior while minimizing detectable geometric deviation. I will investigate whether the distinction between clean and manipulated trajectories remains stable when the attack is adapted to evade the detector. If an attacker can consistently conceal the internal deviation while preserving attack success, the proposed geometric signal should be regarded as insufficient for reliable defense.

Q3. Can geometry-guided intervention causally redirect an unsafe trajectory?

Detection alone does not establish that the identified internal structure plays a functional role in model behavior. I will therefore investigate whether targeted intervention on hidden representations can restore the original task, reduce the influence of injected instructions, or prevent an unsafe external action. This hypothesis would be supported only if the interventions produce predictable improvements in security while preserving performance

under benign conditions. Otherwise, the identified geometry may be correlated with task drift without being causally involved in it.

These questions connect the theoretical idea of input-dependent geometry to a concrete security objective. The purpose is not simply to create a new visualization of latent space, but to determine whether internal geometry can distinguish legitimate context formation from adversarial manipulation, remain informative when an attacker attempts to conceal the manipulation, and support causal control of model behavior.

2.3 Safe Intervention and Human Oversight

Language models and agents may receive a trusted instruction from a user while simultaneously processing untrusted documents, emails, webpages, messages, or tool outputs. If a model internally reinterprets external content as a command, it may depart from the user’s objective before this deviation becomes visible in its final response or external behavior.

I aim to use hidden-state geometry to identify when this reinterpretation begins. A persistent change in trajectory, the breakdown of relationships formed under the original instruction, or the emergence of a new instruction-like direction may indicate that external information is taking control of the model’s task. Unlike a safety mechanism that evaluates only a completed response, this approach would examine the process through which the unsafe decision is being formed.

Detection should lead to an appropriate re-decision. In my Driver Handover research, I studied a system in which decision authority should return to the driver when the state of the human or the environment makes autonomous operation unreliable. I aim to extend this principle to AI agents. When an internal trajectory becomes uncertain or adversarially manipulated, the system may steer its representation toward the original task, withhold a tool call, request clarification, isolate untrusted content, or return the decision to a human.

My legal studies have also shaped this perspective. As AI systems gain access to personal information, external tools, and consequential decision-making authority, meaningful human oversight should not be treated as an interface added only after deployment. It should be supported by the model’s ability to reveal when its own internal process has become unreliable and to preserve opportunities for human intervention before an irreversible action occurs.

2.4 Expected Contributions and Research Direction

This research will develop geometry-aware white-box methods across both state-space models and Transformer-based systems. State-space models will serve as a controlled setting in which sequential dynamics are explicitly represented, allowing the theoretical framework to be formulated and examined clearly. Language models and agents will not be deferred to a distant extension; they will provide an early and essential setting for examining instruction–data separation, prompt injection, task drift, and adaptive attacks.

I aim to make three main contributions. First, I will develop a theoretical framework that treats internal representations as trajectories evolving through input-dependent geometry. Second, I will construct interpretable signals that reveal when trusted instructions, external evidence, and adversarial information begin to alter the model’s internal task representation. Third, I will develop and evaluate white-box control methods that redirect unsafe trajectories, block risky actions, or return decision authority to a human.

Ultimately, I aim to develop AI systems whose decision formation can be observed, experimentally tested, and controlled. Such systems should not merely explain a completed answer after the fact. They should reveal when and how their internal reasoning begins to depart from a trusted objective, remain robust when an attacker attempts to conceal that departure, and support intervention before the deviation results in harmful action.

References

- [1] Sahar Abdelnabi, Aideen Fay, Giovanni Cherubin, Ahmed Salem, Mario Fritz, and Andrew Paverd. Get my drift? catching llm task drift with activation deltas. In *2025 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, pages 43–67. IEEE, 2025.